

Study Questions: Perception — Reading Curvature in the Fabric

1. Pick up a crocheted flat circle and a crocheted piece that ruffles. With your eyes closed, can you tell which is which? Describe the tactile difference between zero Gaussian curvature and negative Gaussian curvature in your own words.
2. Daina Taimina crocheted models of hyperbolic planes to make abstract mathematics tangible. Why is crochet uniquely suited to modeling negative curvature? What is it about the stitch-by-stitch construction that allows the crafter to embed hyperbolic geometry in yarn?
3. Explain the analogy between a CNN's convolutional kernel sliding across an image and a crocheter's fingertip sliding across fabric. What local features does each detect, and how do those local measurements build toward a global understanding?
4. A flat crochet circle requires a specific rate of increase (six stitches per round in single crochet). What happens to the curvature if you increase too slowly? Too quickly? Describe both outcomes in terms of Gaussian curvature and how they feel under the hand.
5. In a convolutional neural network, the first layer typically detects edges. What is the crochet perception equivalent of an "edge" in fabric? What is the most basic, local feature that a crocheter's fingers detect?
6. Describe the four-level hierarchy of crochet perception (stitch, row, section, shape) and map each level to the corresponding feature detection layer in a CNN (edge, texture, part, object). Where does the analogy hold strongest? Where does it stretch?
7. Active Inference describes perception as bidirectional: top-down predictions meet bottom-up sensations. Give a concrete crochet example where the crocheter's top-down belief about the finished object shapes their perception of the current stitch.

8. You are crocheting a hat in the round. The crown is supposed to cup gently into a dome (positive curvature), but it is lying flat. What is the topological prediction error here? How would you correct it?
9. A beginner crocheter finishes a circle that is badly ruffling but says, "It looks fine to me." An experienced crocheter immediately sees the problem. Using the concept of generative model precision, explain why the beginner does not notice the error.
10. Compare topological prediction error in crochet to the loss function in a neural network. In what ways are they similar? In what ways do they differ?
11. When you block a finished piece, you soak it, pin it to a target shape, and let it dry. Describe blocking as a form of free energy minimization. What is the "free energy" being minimized, and what are the constraints?
12. An experienced crocheter can "read the work" — look at a half-finished piece and know what it needs next without consulting the pattern. What level of the perceptual hierarchy are they operating at? How does their generative model make this possible?
13. You crochet a Mobius strip by giving your foundation chain a half-twist before joining in the round. When you run your finger along the surface, you discover it has only one side. Describe this perceptual discovery in Active Inference terms. What prediction was violated?
14. Why does crocheting in the round produce different topological perception than crocheting flat (back and forth in rows)? Consider how the crocheter's ability to see and feel the overall shape differs in each case.
15. A CNN trained on photographs of crocheted fabric might learn to detect stitch types, gauge variations, and shaping. How does the network's feature extraction compare to the crocheter's? What can the crocheter perceive that the network cannot, and vice versa?
16. Describe the sensory experience of discovering a twisted foundation chain after you have worked several rows. Why is this topological error so hard to fix, and why do experienced crocheters check for twists before joining?

17. Precision weighting in Active Inference means that some sensory signals are trusted more than others. When assessing curvature in fabric, which sensory channel — vision or touch — do you think carries higher precision? Does this depend on the type of curvature being assessed?
18. A crocheter is making a sphere (an amigurumi ball). They increase for the first half, then decrease for the second half. At the equator, the fabric should be at maximum circumference with zero local curvature change. How does the crocheter perceive the transition from increasing to decreasing? What signals indicate that the equator has been reached?
19. Neural networks suffer from "overfitting" — learning the training data too precisely and failing to generalize. Can a crocheter "overfit" to a particular pattern or yarn? What would overfitting look like in crochet perception?
20. Reflect on a time when your perception of a crocheted piece changed — perhaps you did not see an error at first, but later it became obvious, or a fabric you thought was wrong turned out to be correct after blocking. Describe this experience as a revision of your generative model. What changed, and why?